

UCE 2026 BIO 1 MOCK COMPILATIONS

ITEM 1

WAKISHA

A study was conducted in a semi-arid savannah ecosystem to investigate how plants survive under extreme environmental stress. They observed that some parts of the study area experience very high temperatures, intense sunlight, low soil moisture, high rates of evaporation and sandy soils with low nutrient content, particularly nitrogen.

In this environment, several plant species such as acacia (leguminous plant) were thriving. Some plants had small needle-like leaves, while others had thick fleshy stems. A number of plants showed deep tap-root systems, while others had widely spread shallow roots close to the soil surface. The legumes had reduced leaf surface area and waxy cuticles; they maintained relatively healthy growth despite the harsh conditions.

The scientist observed that many plants had stomata that remained closed during day but opened at night, and some leaves showed a folded structure during peak heat hours. The ecologists were interested in how the plant species managed to adapt and compete successfully in such a challenging environment.

Task

- a) Identify the plant structures whose functioning was being affected by the environmental conditions in this semi-arid ecosystem
- b) Explain how the environmental conditions in the scenario affected the process of the plants.
- c) Explain how the acacia plants were able to survive and compete successfully with other plants under these conditions.

UTES

John intercropped maize with beans in his garden. The garden was destroyed by a hailstorm at flowering stage, damaging various structures of the maize shoot system leading to losses. However, the bean plants that were growing under the shade of the maize were not destroyed but developed tendrils at the nodes, had long stems, broad leaves and many flowers. The bean plants gave high yields.

Task

- (a) Explain how the hailstorm affected the maize plants causing losses.
- (b) Why were the bean plants able to give high yields?

JJEB

A traditional herbalist collects herbal medicine from medicinal plants in a nearby forest. In the process of harvesting, he frequently plucks leaves and flowers, digs out roots, and strips bark from tree stems. Over time, the population of many medicinal plant species in the forest has greatly reduced. However, one particular species, which has numerous buds and a high capacity for regeneration, has continued to survive and remain abundant.

Task

- a) Explain how the herbalist's harvesting methods affect the normal functioning of medicinal plants.
- b) Explain why one particular species has continued to survive despite the poor harvesting methods.

UTEC

During the rainy season in a certain village in Namisindwa, prolonged heavy rains caused flooding in a low land garden. Most plants showed signs of wilting, yellowing of leaves, and rotting of roots and many eventually died. The excess water in the soils displaced air, creating waterlogged conditions. In contrast, the cypress papyrus plants survived and continued to grow vigorously. On examination, it was found out that this plant had spongy tissue in its roots, its leaves had reduced surface area and had large rhizomes.

Task

- a) Explain how excess water in the soil affected different plant process leading to the death of most plants.
- b) Explain how the papyrus was able to survive in waterlogged soils?

UMTA

A farmer intercropped beans with yams in a wetland without proper drainage channels. The crops grew well at first, however, later the beans developed yellow leaves, weak stems and many had rotten roots leading to no yields. The yams on the other hand, whose leaves had thin cuticles, spongy stems and numerous adventitious roots continued growing normally. The beans barely gave any yields, unlike the yams.

Task

- a) Explain how the conditions in the wetland affected the processes in the plants leading to no yields?
- b) How did the yams in the same wetland garden manage to continue growing normally.

PEA

Lazarus grew beans and Amaranthus dodo in his rocky garden for purposes of food and herbs just during a short spell of heavy rains. The bean plants grew rapidly with large green leaves while amaranthus plants grew slowly and developed relatively small leaves. Just before flowering, a dry spell characterised by hot sunshine hit the village causing the leaves and flowers of the plants to wilt and bending of the stems of the bean plants although the hard amaranthus stems remained erect. During early morning and late evening hours, Lazarus would harvest some upright leaves of amaranthus for food and also dug some of their elongated tap roots for herbs but barely obtained any seeds from the wrinkled bean pods.

Task

- a) Describe how the conditions experienced affected the functioning of the bean plants causing unexpected yields.
- b) Explain how the amaranthus plants managed to survive the conditions experienced to give some yields for his harvest.

MASAKA

Cassava is common staple food in Uganda and beyond. Robert has been a cassava farmer for a long time. Last season, he noted that his cassava plants didn't establish themselves well before the vegetative stage. During the vegetative stage, there was heavy rainfall, which caused waterlogging in the garden. He noted stunted growth characterised by curled, reduced leaf size and yellow patches on some leaves. When the water disappeared, the plants regained their green colour, developed more branches and leaves, which enabled production of some yields.

Task

- a) Explain how the conditions experienced in the garden affected cassava productivity.
- b) Explain how the plants managed to produce some yields when normal conditions returned.

KAMSSA

In the dry season, a wildfire swept through a small forest in a village located in Mityana District. The fire destroyed the apical buds, foliage and barks of most of the trees and shrubs. Several months later, many plants had not recovered because they were completely killed by the fire.

However, *Lophira lanceolata* was observed producing new, healthy green shoots from the ground level shortly after the rains returned. Further investigation revealed that this species possesses; a thick, woody underground tuber which contains stored food reserves and dormant buds, thick-corky bark on the main stem, and its seeds have a hard seed coat cracked by fire.

Task

- a) Describe how the bushfire disrupted key physiological processes in plants, resulting in their death.
- b) Explain how the three adaptations of *Lophira lanceolata* enable it to survive and regenerate after being affected by the fire.

SEEB

Okello is a cassava farmer in Tororo district. Recently, there has been an increase in emissions of sulfur dioxide from factories near his garden. The emissions mix with rainfall forming acid rains. The acid rains burnt and destroyed the bark of the stems, the spongy and palisade mesophyll layers of the cassava leaves. Later at the harvesting stage, Okello found that all his cassava plants had small sized root tubers. To his surprise some nearby trees with thick-bark, thick waxy cuticle and very active shoot meristems survived these acid rains.

Task:

- a) Explain how the acid rains affected plant processes in cassava plants leading to formation of small sized root tubers.
- b) Explain how the nearby trees were able to survive the acid rains.

ACEITEKA

Commercial legume farmers were greatly affected by prolonged drought conditions, which made their crops become stunted, have weak stems and their leaves turn pale green. The farmers were advised to irrigate the crops and when this was done, the crops appeared erect with greener leaves.

Task

- a) Explain how
 - (i) The observations made about the legumes crops came about during such conditions.
 - (ii) The conditions affected functioning of the crops
- b) Describe how the crops utilised the farmers' advice to overcome effects of the conditions.

KEEBS

Mr. Lwanga is a vegetable farmer in a village near a busy stone-quarrying site in Nakasongola District. Recently, he has noticed that his cabbages are growing very slowly and have turned a yellowish-green colour. He also observed that even after a light rain, the plants appear to wilt by midday. A close inspection reveals that the surfaces of Mr. Lwanga's cabbage leaves are covered in a thick layer of white stone dust from the quarry. Additionally, the soil in his garden has become very compacted due to heavy machinery moving nearby, which he suspects is affecting how the roots behave. The quarrying was stopped for one month for a holiday, fortunately it rained heavily on Lwanga's garden, this made him to weed his garden, cabbage leaves started turning green and he managed to realize some yield.

Task

- a) Explain the effect of quarrying on Lwanga's cabbage.
- b) Explain how the cabbage managed to yield after quarry holiday.

ERETA

The primary six pupils of a local primary school decided to grow spinach (*SPINACIA oleracea*) and Sukuma wiiki (*BRASSICA oleracea*) in a large eucalyptus tree garden. One month later, there was a lot of rainfall, days were humid, cold and with very little sunshine for several hours each day. The pupils observed that the vegetables had very small pale-yellow leaves, long thin stems, with wide spacing between nodes. The pupils could not harvest anything and so got worried. Two months later, the rains subsided and the trees were cut down. Pupils applied manure, the vegetables revived growing large thick leaves which pupils harvested and made a lot of money.

Task:

- a) Explain the observed growth behaviour of the vegetables in the first month.
- b) Explain what made the vegetables to revive and yield to the expectations of the pupils.

SEK

Mr. Okello has two fruit trees, a jackfruit and guava planted in the same area on his compound. The Jackfruit tree has many broad leaves, while the guava has many small and slender leaves. During prolonged dry spells, the jackfruit tree sheds most of the leaves, leaving only the young sprouting leaves at the buds. Some of its remaining leaves curl, with most stomata close during the day but open in the night. In contrast, the guava retains all its leaves, remains green and continues to bear fruits throughout the dry season. The jackfruit tree only resumes producing fruits after the rains return.

Task.

- a) Explain how the adaptations shown by the jackfruit tree during the dry season;
 - i) enable it to survive.
 - ii) affect the rate of photosynthesis.
- b) What advantage does the guava have over the jackfruit plant as in the scenario?

ASSHU BUSHENYI

A farmer had a mango tree in his farm on which he would always tether his cow at the tree stem. After sometime the bark at the point of tethering rot away, and there was a swelling on the upper part of the rotten tree bark. During the long dry season, the mango tree almost died. During the rainy season however, a new tissue developed around the point of tethering at the stem, mango leaves became greener and the tree produced more lateral buds.

Task

- a) Explain how the contents in the swollen part of the tree stems were formed within the mango tree.
- b) How did the mango tree survive the conditions it experienced during the rainy season?

KAMTEC

Mr. Mbago transplanted healthy coffee seedlings into a garden during the dry season, expecting rain soon, and also sowed bean seeds in the same garden at the same time. He chose not to irrigate the garden. After two weeks without rain, the coffee seedling leaves turned yellow, their terminal buds dried, and the outer layer of stem also dried, causing the seedlings to become stunted. The bean seeds did not germinate in the dry soil. When the rains finally started, the bean seeds germinated and produced seedlings with healthy shoots and well-developed root systems. The coffee seedlings resumed and new shoots developed around the nodes. The coffee and beans managed to give good yields.

Task

- a) Explain how
 - (i) Transplanting the coffee seedlings during the dry season affected them
 - (ii) The coffee seedlings and bean seeds managed to behave the way they did when rain started.
- b) Explain how the crops managed to survive and give some yields at the end of the season

KASSPHA

Students in the environment club carried out a tour around their school compound and noticed that some trees were damaged. They decided to monitor the trees over time in order to find ways of saving them. They collected the following information:

Tree	Type of damage	Effect of the damage
Jackfruit	The bark was removed making a ring around the stem	-Leaves dropping and drying -Swelling of the upper edge of the cut surface - Tree dying
Mango	- caterpillars feeding on leaves -part of the bark was removed	-Swelling around the cut area -Stunted growth
Orange	Attacked by a fungus that blocks the xylem	Severe wilting with dropped leaves

To save the trees, all of them were watered and supplied with fertilizers. In addition:

- The mango tree was sprayed with an insecticide to kill the caterpillars.
- The orange tree was sprayed with a fungicide to kill the fungi. After two months, the effects of the mango and orange trees were reversed and they produced some fruits, but the jackfruit tree died.

Tasks.

- a) Explain how the type of damage in each tree led to the observed effects.
- b) Explain why the mango and orange trees were able to survive and produce fruits.

LUDISSEB

Kabanda planted beans in his piece of land located in a valley. The plants had green leaves and had many flowers. Due to stagnant water in the garden, the leaves of the bean plants turned yellow, many flowers withered and the roots became rotten leading to death of most of the bean plants. However, the cocoyam plants in the same piece of land with numerous adventitious roots, broad leaves with thin cuticle, and large air spaces in their stems survived.

Task

- (a) Explain
 - (i) How the observations made about the bean plants came about as a result of the stagnant water.
 - (ii) How the stagnant water affected the normal functioning of Peter's bean plants.
- (b) How did the coco yams manage to survive the conditions in the garden?

Item 2

JJEB

An old woman has been suffering from joint pain and frequent fractures. Her son developed visual and hearing impairment. When the son was diagnosed in hospital, the clinical laboratory report indicated that the auditory canals had a lot of wax and the ear drums and cochlea were damaged. The iris had less pigment and the ciliary muscles were weak. While in the hospital, the old woman was informed that her sickness was due to hormonal imbalance as a result of her old age. The woman is so worried about her health condition.

Task

- a) Explain the cause of the son's impairments as indicated in the report.
- b) Explain what can be done by the old woman to manage her health condition.

KASSPHA

Kevin, a 35-year-old long-distance truck driver, was hospitalized with multiple fractures after a road accident caused by a sudden episode of blurred vision. In the hospital, laboratory tests revealed glucose in his urine. Kevin admitted to frequently using a stimulant drug to stay awake during long journeys. Subsequent evaluation revealed that the drug suppresses insulin production, damages the optic nerve, and inhibits deposition of calcium in bones.

Task

- a)
 - i) Describe how the stimulant drug caused Kevin's accident.
 - ii) Explain Kevin's multiple fractures and his lab results.
- b) Advise Kevin on how to manage his life to ensure better health.

LUDISSED

As James was running away from police, he accidentally stepped on a sharp thorn and suddenly folded the leg at knee joint, and due to pain, lowered it as he limped. James has a short stature with a swelling in the neck region and these affect his life. When he went to hospital, he was informed that his health challenges were due to insufficient hormone secretion.

Task

- (a) Explain how Page 6 of 34 Compiled by Alinda Justus
 - (i) James' body coordinated to bring about limping suddenly after stepping on the thorn.
 - (ii) Structures in the leg worked together leading to limping
- (b) Explain the causes and management of the health challenges affecting James's life.

UTES

Mary and Joseph have been married for 20 years and have four girls and one boy. When they produced the fifth girl, Joseph decided to abandon the family and started having unprotected sex with other women in quest for more boys and so he became HIV positive. Their eldest boy had colour blindness, which is affecting his performance at school. Joseph finally produced a baby boy outside his marriage, however, the child had Down's syndrome.

Task

- (a)
 - (i) Genetically show how their eldest son inherited the condition that is affecting his performance at school.
 - (ii) Explain how the condition present in the boy born outside Joseph's marriage came about
- c) Explain how
 - (i) Joseph can manage his health status.
 - (ii) The condition in the boy child born outside marriage can be managed.

WAKISSHA

In Lufunda Parish, a rural farming community, the local healthy centre has reported a rising number of chronic health conditions affecting both adults and youth. During a medical outreach program, health workers observed the following; -

- Many adults especially those above 45 years, are suffering from weak bones, frequent fractures and persistent back pain (osteoporosis).
- many women and children were having visible swelling at the front of the neck (Goitre)
- Several middle-aged individuals were diagnosed with diabetes mellitus.
- Many youths frequently consume tobacco, alcohol and other substances.
- Some patients also showed reduced sensitivity to environmental stimuli such as light and sound to prolonged substance abuse.

The medical team confirmed that the conditions were linked to hormonal imbalance, enzyme inefficiency, poor diet and substance abuse affecting their body systems. Clinical tests confirmed diabetes, goitre and osteoporosis.

Task

- a) Explain how hormonal imbalance contributes to the development of diabetes, goitre and osteoporosis in the individuals of Lufunda Parish.
- b) Explain how reduced enzyme activity affected absorption and overall health of the people in Lufunda Parish.
- c) Discuss the role of diet and lifestyle in the prevention and management of the disorder identified in the community.

UTEC

Mugerwa, a S.4 student in Wakiso district, has been experiencing fatigue, extreme thirst and blurred vision. At the hospital, urine dialysis test showed glucose present. The doctor diagnosed him with a condition affecting blood sugar control and advised him to manage his diet to live a health life. One day at school in a P.E lesson, Mugerwa repeatedly raised and lowered his arm to throw a javelin as part of an exercise, but during the drill he saw a big snake in the bush. He became frightened, his heart beat rate increased, his pupils dilated, he sweated excessively, and he ran away.

Task

- a)
 - (i) Explain how Mugerwa's conditions resulted in the health challenges he experienced before seeing the snake.
 - (ii) Show how Mugerwa's body coordinated to bring about the fear responses when he saw the snake, and also to control the movement when raising and lowering his arm.
- b) Suggest practical ways Mugerwa can implement the doctor's dietary advice to manage his condition and improve his ability to carry out daily activities.

UMTA

Mr. Mulunga, a 60 year old man became ill and was hospitalised. Medical examinations revealed two health conditions; high blood sugar levels and osteoporosis, both attributed to old age. When in the hospital, doctors put him on a therapy (regular light exercise) involving repeated folding and straightening of his leg aimed at reducing effects of the conditions.

Task

- a) Explain,
 - (i) The cause of the health conditions revealed in Mr. Muluga, apart from old age.
 - (ii) How structures in the legs worked together to bring about the observed actions during the therapy.
- b) Apart from regular exercise and diet, explain medical interventions to manage Mr. Muluga's health conditions.

PEA

Kiplimo 35-year-old long-distance runner was recently diagnosed with a medical condition that causes weak and brittle bones. His heart beat increases shortly before starting whistle for the race is blown. Kiplimo managed to start the race well. However, during the final stage of the race, he experienced severe muscle cramps in his left leg. He attempted to jump using the right leg in order to reach the finishing but unfortunately, his femur bone became fractured and the knee joint was injured in the process, which made him unable to reach the finishing line.

Task

- a) (i) Describe how Kiplimo's body coordinated to enable him to start his race well.
 - (iii) Explain how challenges he experienced in his final stage of the race made him unable to reach the finishing line.

- b) Advise Kiplimo on how he can manage the health condition that caused him weak and brittle bones.

KAMTEC

Obote was persuaded to start smoking with his friend's marijuana in the nearby trading center, which he is now addicted to. One day during a police hunt, he accidentally stepped on a sharp thorn as he was running away and immediately limped before he could stop to remove it. Although he successfully removed the thorn from his foot, the wound could not easily heal due to a health condition brought about by his faulty pancreas. John also passes out small volumes of concentrated urine and complains of thirst all the time.

Task

- a) Explain,
 - (i) how Obote's reaction came about when he stepped on a sharp thorn.
 - (ii) The cause of the health conditions experienced by Obote.
- b) How can John manage his health conditions to live a better life.

MASAKA

A 16-year-old girl in S.3 enjoys being in the company of boys who were suspended from school over drinking and sniffing petrol. Her parents noted decline in her performance over the last term, when the parents inquired, she complained of a persistent headache during class time. The mother gave her some tablets to swallow, which she did. Unfortunately, she vomited the tablets after swallowing. The mother got worried and took her to the hospital. On medical examination, the doctor realised that the girl has been using strong drugs without eating and had contracted HIV. This has contributed to her weight loss, poor memory retention, nausea and increased blood pressure.

Task

- a) Describe how her response to tablets swallowed came about.
- b) Suggest suitable advice to improve her condition and academic performance.

KAMMSA

Okedi is a boda-boda man in Kumi district. Every evening he rides his motor bike to an alcohol drinking joint. One day after taking five bottles of alcohol, he left the drinking point when he was staggering, could hardly remember anything and had delayed responses. As he was riding back home, he crashed into a wall and he was rushed to the hospital. To determine the extent of damage to his nervous system, the doctor tapped just below his kneecap with a tendon hammer and observed that his leg kicks.

Task

- a) Explain how alcohol affected Okedi's nervous system leading to the situation he was in when he left the drinking point.
- b) Describe the sequence of events involved in producing Okedi's response when the doctor tapped below his kneecap at the hospital.
- c) How can Okedi avoid challenges associated with his habit?

SEEB

Mr. Kale is a 50-year-old biology teacher who suffers from a health challenge of Type II diabetes. One day while inspecting a school assembly, he stood with his legs crossed near a sports field. A student accidentally kicked a small stone which travelled and hit the teacher sharply below his kneecap of one of his legs. Before the teacher could even turn his head to look, his foot swung forward in a sudden kicking motion. The teacher was surprised on how fast his body coordinated to bring about such a response.

Tasks:

- a) Explain how Mr. Kale's body coordinated to bring about the response observed in the image.
- b) Suggest lifestyle strategies Mr. Kale can undertake to manage the health challenge he is suffering from.

ACEITEKA

An old woman previously diagnosed with diabetes suffers with blurry vision. While looking at her flowers at home, her eye was approached by a speeding insect, making her immediately close it and turn away. Because of her health condition, her feet periodically swell and she has poor wound healing.

Task

- a) Explain
 - (i) How the old woman's reaction towards a speeding insect came about.
 - (ii) How the health condition resulted into the challenges she is experiencing.
- b) Advise the old woman on how she can manage her health condition.

KEEBS

Okello is a talented Senior Four student who recently joined a local community football team. To keep up with the older players, his friends introduced him to "Khat" (Mairungi) and "Kuber," claiming these substances would give him "limitless energy" and keep him alert during matches without training. However, during a high-stakes regional tournament, Okello's performance took a worrying turn. Ten minutes into the second half, Okello's movements became uncoordinated, complaints of thirst, and increased heartbeat. Suddenly, he collapsed on the pitch crying his left calf muscle had become hard, tight, and extremely painful. He had to be carried off the field, unable to continue. The team medic noticed Okello was severely dehydrated and trembling.

Task

- a) Explain the cause of Okello's experience during the game
- b) Advise Okello on how he can manage the condition that he is experiencing

ERETA

Mandu works at a construction site where he overstrains his muscles, stands for long hours, drinks less water but a lot of alcohol and smokes shisha late in the evening. Over time, he started feeling muscle pains, numbness, paralysis, poor body coordination and irregular heartbeat. When the doctor examined him by tapping his knee, his lower limb kicked forward but with less vigour indicating that his muscles had weakened.

Task.

- a) Explain how the conditions experienced by Mandu could be due to the nature of his work and behaviour.
- b) Describe the events that led to his leg to kick forward.
- c) How can Mandu prevent the conditions experienced in his body?

SEK

At Pima Secondary School, three learners were interviewed about changes observed in their bodies under different conditions. Their responses were recorded as follows:

- Learner A: Reported difficulty in seeing, frequent thirst, hunger and frequent urination.
- Learner B: Reported difficulty in seeing writings on the chalkboard clearly, although writings in books could be seen clearly.
- Learner C: Experienced excessive sweating, rapid heartbeat and heavy breathing after being caned by a teacher.

Task

- a) Identify the conditions experienced by learners A and B.
- b) Explain how the conditions experienced by learners A and B can be corrected.
- c) Describe the body response shown by learner C after being caned.

ASSHU BUSHENYI

Francis is a teenager who dropped out of school due to his involvement in a group of boys in town who drink alcohol and smoke marijuana. He spends most of his time with the group and they are known for crime in town. One day there was a demonstration in town and the police fired teargas at people. When Francis heard the loud sound of the tear gas canister, his heart pounded, eyes protruded and he urinated on himself. He tried running but his legs were too weak to support him.

Task

- a) Explain
 - (i) The effects of Francis' lifestyle.
 - (ii) How his body coordinated to bring about his reaction after the tear gas canister.
- b) Advise Francis on how to overcome the negative effects of his life style.

ITEM 3

JJEB

A 15-year-old Senior Three homozygous normal girl became pregnant after engaging in a sexual relationship with a heterozygous normal man from a family with some albino members. The pregnancy has negatively affected her. She is also worried that the unborn baby may be albino because albinism exists in the man's family. Due to this and the challenges associated with the pregnancy; she is considering abortion. However, her friend has advised her against making that decision.

Task

- a) Genetically show to the girl the possible phenotypes of the unborn child.
- b) Explain the likely challenges that the girl could be experiencing due to her current condition.
- c) Explain the importance of the friend's advice to the girl.

KASSHPA

David got married to Joan and they produced a son. Their son, when injured, bleeds a lot even with a slight cut and this causes a lot of distress in the family. When Joan got pregnant for the second time, David advised her to abort on the belief that they would get a child with similar condition as that of their first-born son. However, when David and Joan went to the hospital concerning their son's condition, they were told that the condition was a result of Joan being a carrier of a bleeder's disease.

Task

- a) Explain why Joan should not take David's advice.
- b) Show how their son inherited the condition.

KAMTEC

Mr. Lutaaya is a maize farmer in Luwero district. He consistently faces significant challenges due to a common fungal disease that severely affects his local maize variety, leading to yield losses and impacting his income. He recently learned from an agricultural expert that disease resistance is a dominant trait in maize over disease susceptibility trait. He managed to acquire some seeds from a special maize plant that is pure breed for disease resistance and he has a dream of having disease resistant plants in his garden but he is not certain of how to achieve it.

Task

- a) Explain to Mr. Lutaaya how he can achieve his dream of having disease resistant maize plants in the second generation offsprings. (*use correct genetic symbols to show how he can achieve it*)
- b) Describe the significance of cultivating disease-resistant maize to Mr. Lutaaya's farm and the local community.

WAKISHA

A couple living with HIV has three children with normal vision, however the fourth child, a boy was found to have difficulty in distinguishing between red and green colours. This caused concern to the parents since neither of them had any noticeable vision problems. After medical examination, the doctor diagnosed the child with red-green colour blindness, a genetic condition linked to the X-chromosomes. The doctor re-assured them that the child could live a normal life with proper support.

Task

- a) Using genetic symbols, show how the conditions in their fourth child came about.
- b) Explain how the couple can;
 - (i) Continue living positively with their health status.
 - (ii) Take care of their fourth child.

LUDISSEB

Daniel and Rose produced a baby. The baby has swollen and painful joints, frequent tiredness, shortness of breath and dizziness. When Rose got pregnant again, Dan advised her to abort thinking that they would get a child with similar condition. However, when Dan and Rose went to Hospital concerning their baby's condition, the doctor told them that, the health challenges were due to the couple being heterozygous for a genetic condition.

Task

- (a) Explain how
 - (i) The genetic condition resulted into the health challenges in the baby
 - (ii) The baby inherited the genetic condition from the parents.
- (b) How can Daniel and Rose manage the genetic condition of their baby to ensure better life?

UTES

Mary and Joseph have been married for 20 years and have four girls and one boy. When they produced the fifth girl, Joseph decided to abandon the family and started having unprotected sex with other women in quest for more boys and so he became HIV positive. Their eldest boy had colour blindness, which is affecting his performance at school. Joseph finally produced a baby boy outside his marriage, however, the child had Down's syndrome.

Task

- (a) Genetically show how their eldest son inherited the condition that is affecting his performance at school.
- (b) Explain how Joseph can manage his health status. (ii) The condition in the boy child born outside marriage can be managed. (i) Explain how the condition present in the boy born outside Joseph's marriage came about.

UTEC

Reproductive health Uganda (RHU) is a community-based organisation in Kampala District. It supports young people to make informed health choices. RHU works through school health clubs and community outreaches with three main focus areas;

Mandate 1; educate adolescents about reproductive health, with special attention to the menstrual cycle as a key change during puberty in girls.

Mandate 2; create awareness on Sexually Transmitted Infections, particularly HIV/AIDS, and encourage practices that promote healthy living.

Mandate 3; sensitize families on inherited conditions.

Recently, RHU met the Mukasa family. Mr. mukasa has a normal colour vision, but his two sons cannot distinguish between red and green colours. The doctor explained that this is a genetically inherited disorder.

Task

- a) Explain how the puberty process in mandate 1 above takes place in adolescent girls.

- b) (i) Using the Mukasa family as an example, explain how the disorder was passed on to the sons
 (iii) Advise adolescents in the community on how to address the challenges highlighted in mandate 2 of RHU.

UMTA

A couple was blessed with five children born in three births, each separated by two years. The children were of different sexes, physical appearances and showed a number of variations. The table below summarises notable features about the children.

Pregnancy/birth number	status	sex	Physical appearance	Degree of variation
1st	Twins	Boys	Identical	Slight
2nd	Twins	Boys	None identical	Much
3rd	Single baby	Girl	Flat face Short neck Small ears	Very much

Task

- a) Explain how,
 (i) The 1st and the 2nd pregnancy/birth numbers with such physical appearance arise with in the mother's womb?
 (ii) The much variation came about in the 2nd birth number/pregnancy?
 b) Advise the couple on how to take care of their third born child.

PEA

Enock got married to Edith and produced a pair of twins, a boy and a girl. Both twins experienced swollen and painful joints, frequently fall sick and their skin and eyes appear yellowish. When Edith got pregnant again, Enock advised her to abort thinking that they would get another child of similar condition. However, when Enock and Edith went to hospital concerning their twins' condition, the doctor told them that the health challenges experienced by the twins were due to the couple being heterozygous for sickle disease.

Task

- a) (i) Using a diagram, explain how the twins inherited the genetic condition.
 (iii) Explain the dangers Edith is likely to face if she follows Enock's advice.
 b) How can Enock manage the genetic condition of his twins to ensure better life.

MASAKA

A fifteen-year-old girl got pregnant, and her parents forced her to marry the man she claimed was responsible for the pregnancy. The man was found living in a slum area with poor feeding and excessive drinking of alcohol. This troubled the girl, and she failed to attend antenatal care services. After 9 months, she delivered an underweight, pale-skinned baby with white hair and pink eyes. The man was shocked and denied being the father of such a child. Further investigations with DNA confirmed that he is the biological father of the baby.

Task

- a) Using genetic symbols, explain to him that it is possible to have such a child.
- b) Explain
 - (i) How girl's condition contributed towards having an underweight child.
 - (ii) Why is knowledge of genetics essential for this family?

KAMMSA

Mercy and her husband had been taking their baby for routine immunization and medical care. The nurses kept on plotting the baby's weight curve on the immunisation card. On one visits, the nurses got concerned and referred the baby to the doctor due to very low growth rate, pain crises, swelling at the joints and fingers. The doctor diagnosed the baby with a genetically inherited disease.

Task

- a) Identify the genetic disease and how it was genetically passed on to the baby.
- b) Describe the likely signs and symptoms of the baby.
- c) How the family can manage their baby's condition?

SEEBs

Recently Sally a resident of Kamwenge got married to Ibrahim. They produced their first son Max who is a Sickler. Currently Sally is expecting a daughter and the family is worried of having another child who is a Sickler due to the challenges their son Max experiences. The below shows part of Sally's family tree where some members of her family have sickle cell anaemia. A year later after giving birth, Sally visited a fertility clinic to monitor her reproductive health. Over a period of 28 days, a gynaecologist used an ultrasound transducer to measure the thickness of her uterine lining (endometrium) at regular intervals, producing the data plotted in the graph below.

PICK THE PICTURE

Task

- a) Using knowledge on the hormonal control of the menstrual cycle, explain what is happening at points A, B, C and D. In each case, give the significance of what is happening.
- b) Using genetic knowledge, determine the probability that the daughter Sally is expecting inherited the condition.

ACEITEKA

A couple had no child for many years. They decided to go to hospital and the clinical laboratory results indicated that the wife had blocked fallopian tubes, cysts around her ovaries and damaged inner uterine lining. When the wife received treatment, the couple produced an albino child. However, the husband denied being responsible for the child since the couple lives with an albino man in their neighbourhood. The child has many challenges and this stressed the mother.

Task

- a) (i) Explain how the observations in the clinical laboratory report led to the wife's failure to have children.
- (iii) Using a genetic diagram, explain to the husband the possibility of being the biological father to the child.

- b) How can the challenges associated with the child's condition be managed?

KEEBS

In the rural village of Ajonyi, a recent community health survey revealed that many families are struggling with reproductive health challenges. Robert and his wife, Grace, are expecting their fourth child. Robert has been expressing concern, hoping for a son to "continue the family name," and he has been blaming Grace for their previous three daughters. They were advised to visit a professional counselor who then told them children's sex are determined by the biological father, Robert became annoyed that Grace bribed the counselor to talk against him. Due to terrible words that Robert told his wife Grace, she became unfaithful and started having affairs outside marriage, became pregnant and abandoned going for ante natal care, contracted syphilis and HIV/AIDS. As a result, she gave birth to a baby boy unfortunately the boy was also tested positive for HIV and her husband as well. Meanwhile, the village health team has noted an increase in reproductive health complications among teenagers, including undiagnosed infections and early pregnancy risks.

Task

Using your knowledge of genetics,

- (a) explain to Robert how sex of children is determined by the biological father (
- (b) Explain;
 - (i) the likely effect of Grace's life style to her family
 - (ii) how she can overcome the challenges she is facing and that of her child

ERETA

Josh who had just finished high school made pregnant a 16-year-ld Nambi who despite looking normal, hails from a family in which the father has a challenge of the inability to distinguish between shades of red and blue colours. This has worried the father to josh because Nambi is just a teenager and given her family history, he thinks all his grandchildren might inherit the challenges of this family if at all josh is forced to keep Nambi for marriage.

Task.

- a) Explain
 - (i) why josh's father is worried of Nambi being a teenager in her observed state of health.
 - (ii) using genetic diagrams, the possible phenotypes of the grandchildren to josh's father.
- b) Suggest how to manage the challenge associated with Nambi's family to reduce on the worries of josh's father.

SEK

Achom and Oputiri have been married for several years and have always believed that they were both healthy. However, their son has a condition in which he bleeds for a long of time before the blood clots. During a family counseling session, Oputiri also noticed that his son had become more restless and emotionally unsettled as he passes through puberty.

Task:

- (a) Using your knowledge of genetics, explain to the couple the cause of their son's health condition.
- (b) Suggest ways in which the couple can help their son manage the health condition.

ASSHU BUSHENYI

A couple living with HIV has three healthy children. Their fourth born however was born with pale skin, white hair and pink eyes. This worried them, since they were both dark-skinned but the doctors advised them to take care of the baby properly, since it was a genetic condition and so the baby would grow normally as others.

Task

- a) Using genetic symbols, show how the fourth born inherited the condition.
- b) Explain how the couple can
 - (i) Continue living positively with their health status
 - (ii) Take care of their fourth born child.

SECTION B

ITEMS 4 AND 5

KAMTEC

Mr. Muhindo bought a piece of land along the slopes of a hilly area in Kasese. He cleared a forest to create space for growing maize, burnt the cleared vegetation and also burnt the cut trees for charcoal. For three consecutive years, he has been growing maize on his garden and always applied artificial fertilizers to increase crop yields, and pesticides to eradicate pests. After harvesting, he always cuts down the maize plant residues, burns them and ploughs the area using tractors. These have caused many challenges to the environment and he has registered a decline in yields for the past two seasons.

Task

- a) Explain the likely challenges that may arise from Mr. Muhindo's actions.
- b) Explain why the environment should be sustainably utilized by Mr. Muhindo without much damage.

In Rushesha village, on the slopes of Mount Rwenzori, farmers rear large numbers of cattle and grow food crops, mainly Irish potatoes. In the recent years, the farmers have suffered due to frequent erosion and flooding of their gardens and compounds especially during the rainy seasons, their crops are frequently attacked and eaten by wild rats. Depletion of grazing areas on the mountain slopes, decreased rainfall and increased cases of malaria and cholera have also been observed in the area.

Officials from the district have made an environmental assessment report on the area and have highlighted the following;

- *Heavy reliance on firewood as a sole source of fuel*
- *Most households lack pit latrines and defecate in bushes near their homes*
- *Some locals lay traps to capture snakes for their skins, which they sell for livelihood.*
- *Many pits dug during brick laying get filled with water in the rainy season*

Task

- Explain how the human activities in the area have led to the challenges faced in the area
- Explain ways in which the challenges in the area can be solved.

WAKISHA

Residents of Kayanja village depended on a nearby wetland for clean water, fishing and harvesting papyrus for making mats and baskets. The wetland also supported a wide variety of birds, insects and helped regulate flooding during rainy season. Over time, a real estate developer drained a large portion of the wetland and filled it with sand to construct residential houses. In addition, waste from houses was directed into a small channel that originally fed the wetland. After few years residents began to notice reduced fish populations, frequent flooding during heavy rains, foul smelling water and increase in water borne diseases.

Task

- Explain how the developer's activities lead to the loss of the benefits previously obtained from the wetland by both the environment and the residents.
- Suggest the importance of conserving wetland like the one described in the scenario.

The Uganda Wild Life Authority conducted a study in Murchison fall national park surrounding communities to assess the impact of increasing human activities on wildlife population and ecosystem stability. Over the past five years there has been; expansion of farmland into protected areas, increased poaching of herbivores such as antelopes, charcoal burning and deforestation near park boundaries. At the same time communities have reported rising cases of crop destruction by antelopes, increased attacks on livestock by lions and reduced fish catch from the nearby water bodies.

The figure below shows changes in wild life population fish and human activities (2016-2020)

Year	Antelope popn	Lion popn	Poaching cases	Fish popn
2016	12000	320	80	70
2017	11200	310	110	65
2018	10060	290	150	55
2019	8800	270	210	40
2020	7500	212	260	32

The changes in fish population was linked to industrial discharge and sedimentation from farming.

Tasks

- Explain the effects of human activities on the ecosystem
- Explain how these changes hinder socio-economic development.
- Suggest solutions to the challenges and benefits of implementing the solutions.

UTEC

Mabira ridge forest reserve was once a balanced montane ecosystem (dynamic and highly elevated). Shae and ironwood trees formed the main plant base of the food web. They provided food for large population of forest duikers. The duikers' numbers were kept in check by a healthy population of several cats, the main predators. In the last five years, illegal trapping has sharply reduced the number of several cats. During this same period, conservation biologists working with

local forest rangers monitored key ecosystem indicators. Their records for 2021 and 2025 are shown in the table below.

Indicators	Year 2021	Year 2025
Serval cats	150	5
Forest duikers	5200	700
Numbers of trees (shea and iron wood)	High	Low
Level of top soil nutrients	High	Low
Cases of crop raiding by forest duikers	Low	High

Task

- Explain how the decline in serval cat numbers could have led to the changes recorded for each indicators in the table.
- Suggest practical steps that can be taken to restore the forest ecosystem to its 2021 condition.

River Basin Community, located next to a newly built hydroelectric dam created upstream. To improve household income, residents started large-scale rice cultivation along the dam's edge. The farms depend heavily on chemical fertilizers and raw animal waste. When it rains, washoff and farm drainage flow straight into the reservoir. The dam has two critical roles for the area; it supports the community's main fishing activities, and supplies drinking water to a fast growing market centre downstream. Over five years, a multi-agency team made up of fisheries, environmental and public health officers monitored changes in the ecosystem and human health. Their data for 2021 and 2025 is shown below in the table.

Category	2021	2025
Water clarity	clear	Highly turbid and muddy
Water hyacinth and algae	5%	85%
Dissolved oxygen	Normal	Very low
Tilapia fish population	45000	4200

Task

- Explain how the farming practices described could have led to the changes observed in the dam's ecosystem.
- Suggest sustainable actions the community can take to protect both their livelihoods and the dam's water quality and give the benefits of conserving such biodiversity.

UMTA

A certain city has been face with many challenges due to wetland destruction. People near these wetlands cleared vegetation, pour large volumes of soils in swamps to construct houses and factories, they also dump plastics into the wetlands and release untreated industrial wastes in these wetlands. The authorities attribute the challenges to these practices and recent they decided to demolish/destroy buildings that had been constructed in a large wetland and evicted/chased away all the people who were staying in the wetland area. However, the evicted people were unhappy with the authorities' actions.

Task

- Explain how people's actions caused challenges in the area.
- What could be done to minimize the challenges in the area?
 - Of what importance to man would it be such area were left undisturbed?

After the Kitezi garbage slide over people's homes that destroyed lives and property, the city council resolved to stop dumping garbage in the affected area. This resulted into poor waste management in Kampala City especially in the suburb. People started dumping domestic wastes in nearby swamps, burning plastics and dumping garbage in drainage channels and on streets. As a result, the communities are experiencing several challenges.

Task

- a) Explain the challenges likely to be faced by the city residents if they continue with their way of waste management in Kampala.
- b) Advise the city residents on how to handle the problem without causing damage to the environment.

PEA

Queen Elizabeth National Park covers a vast land area in the western parts of Uganda and its home to millions of organisms. However, in the last years, many people have encroached on the area, cleared trees and bushes for agriculture and settlement, charcoal burning and lumbering. Many have also started dealing in bush meat and elephant tusks for livelihood. Reports from the wildlife authority show a decrease in numbers of elephants and rhinos in the park, frequent migration of birds from the area, decreased rainfall, extinction of traditional tree species in the area, frequent attacks of wild animals on humans and some cases of ebola among the neighbouring communities.

Task

- a) Explain the challenges caused by human activities around the national park on the environment.
- b) Suggest practical solutions to the challenges being faced around the national park.

The government of Uganda through the NAADS program provided cassava cuttings to the residents of Ombokolo to enable them grow their own food. The residents cleared many shrubs and indigenous trees on very large areas on their communal land, in order to mass produce cassava which is their staple food. Many also used different insecticides to destroy ant hills to limit chances of termites destroying the cassava plants as well as applying fertilizers to boost the harvests. However, these actions have resulted into a number of environmental challenges in Ombokolo.

Task

- a) Explain to the residents the challenges they have caused to the environment as a result of their actions.
- b) Why it was important for the residents to conserve their environment amidst the prevailing famine problem.

MASAKA

According to the national population and housing census 2024, Uganda's population increased to 46 million people, up from 14 million in 1986. This is associated with significant demand for human settlement, wood fuel, overgrazing and over cultivation, causing depletion of the green cover. The national environment management authority reports show a significant decline in soil productivity, biodiversity loss, increased soil erosion and increased temperature. This threatens Uganda's vision 2040, which focuses on the sustainable use of natural resources.

Task

- Explain how population expansion threatens the attainment of Uganda's vision 2040
- Suggest measures that can be adopted for sustainable development amidst the rapid population expansion in Uganda.

Aquatic ecosystems provide essential benefits such as food, oxygen and water for sustainable living. However, farmers around the lake Victoria basin threaten these benefits due to increased use of agrochemicals for fish farming and rice growing. Environmental quality data for the last decade in figure 1 show variation of agrochemical use with water quality, fish catch, and health risks around the basin.

Graph

Agrochemical use, water quality, fish catch and health risks (2016-2025)

Task

- Explain how the use of agrochemicals around the basin affects the ecosystem benefits.
- Suggest the strategies that can be adopted by the farmers to ensure sustainable living with nature.

KAMMSA

Kidepo-edge Mahogany Forest was once a stable ecosystem. The foundation of its food web was mahogany and acacia trees, which supported a large number of bush pigs. This herbivore population was naturally controlled by African leopards. In recent years, unlawful hunting has greatly decreased the number of leopards. Over a period of five years, ecologists together with local community leaders monitored the health of the ecosystem and recorded the following data

Indicator	Year 2021	Year 2026
African leopard numbers	45	3
Bush pig numbers	3201	850
Number of trees	306	41
Level of topsoil nutrients	High	Low
Cases of crop raiding by bush pigs	12	245

Task

- Explain how the decline of African leopards caused the changes observed in each indicator shown in the table.
- Recommend measures that can be taken to restore this ecosystem to its 2021 condition.

30 years ago, Lubigi was one of the largest wetlands in Kampala district. In early 2000s individuals have cleared and constructed building and various factories in the wetland which resulted into many environmental problems.

Due to many environmental challenges brought about by the actions, the national environmental management authority *NEMA (has decided to force them out of the area and destroy all their buildings in order to restore the wetland and residents are not happy.

Task

- Explain the environmental challenges that resulted from the individuals' actions in the area.

- b) Suggest what could be done to avoid such environmental challenges if the residents were to live in the area sustainably and why it is important for NEMA, to restore the area to its original state.

SEEB

A group of researchers from a conservation organization conducted a long-term ecological study on a wetland in Uganda. This wetland ecosystem mainly supports a population of Tilapia fish and Fish Eagles (birds), where the fish Eagles feeds on the tilapia fish. During the first six years of the study, the two populations fluctuated naturally in a stable predator-prey cycle, however in sixth year, the local community began intensive sand mining in the wetland and heavy use of inorganic fertilizers in adjacent commercial farms simultaneously. During the rainy seasons, high concentrations of nitrates and phosphates get into the wetland. The changes in the population of the two animals over ten years are shown in the figure below.

Task:

- a) Explain;
- i) the relationship between the population of fish eagles and tilapia fish from the start of the study to 4.5years
 - ii) why activities of the local community caused a decline in the populations of the two animals.
- b) The conservation organization wishes to restore this ecosystem, suggest some actions it can take.

In one garden ecosystem in Mityana district, a bird of prey called a sparrow hawk hunts in a field of clover plants, searching for its prey another bird called a thrush. The thrushes visit the field to feed on caterpillars. The caterpillars graze on the clover leaves. In the same garden ecosystem, there are wild rabbits which feed on commelina a weed in the garden which is also fed on by caterpillars and wild cats which feed on the rabbits, thrushes and hawks. Recently a farmer cleared all the trees in this garden to get timber and sprays the clover field with an insecticide, killing all the caterpillars and this has led to reduced crop yields.

Task

- a) (i) Construct a food web for this ecosystem.
(iii) Explain the effects of the farmers actions on the garden ecosystem.
- b) Suggest sustainable remedies the farmer can apply to restore this garden ecosystem.

ACEITEKA

Mr. Okoli is a known farmer in Kapeka. He bought a large piece of land near the permanent swamp which supplies the community with water and fish for food. When it was nearing planting season, he cleared most of the vegetation to open up a farm land. During the planting season, he sprayed with insecticides to kill insects in the swamp to avoid pests. In that year, the community experienced several challenges which were attributed to Okoli's actions.

Task

- a) Explain the challenges likely to be faced by the community due to Okoli's actions.
- b) Advice Okoli on how he can do his farming activities without harming the community.

Part of a natural swamp was given out to an investor to construct factory. The investor cleared all swamp vegetation, poured large volumes of soil, which he compacted into the swamp before construction began. After sometime the surrounding community complained of rampant floods and drying of natural water sources.

Task

- a) Explain the problems that the investor caused to the natural environment and suggest possible measures to overcome them.
- b) How beneficial is such a natural environment when left undisturbed.

KEEBS

In Aboket village, the local wetland has traditionally provided clean water, mudfish, and a breeding ground for the Grey Crowned Crane. However, over the last 15 years, the village population has grown rapidly. To sustain their families, many residents have turned to large scale sand mining and the conversion of the wetland into rice paddies. The local environment officer has noticed a sharp decline in biodiversity. The water has become dark cloudy and shallow, and the local fishermen are struggling to find enough fish to sell. A census of the Grey Crowned Crane shows that nesting pairs have dropped from 40 to only 4 in the last decade. The community is now facing a shortage of clean water and a loss of their primary source of protein. The table 1 below shows data collected by the environment officer over the years.

Table 1

Year	Human population (thousands)	Monthly fish yields (kg)
2010	1.2	850
2015	2.8	580
2020	4.5	310
2025	6.7	95

Population and fish yield in aboket village

Task

- (a) Using the data provided in the table explain;
 - (i) the relationship between the human population growth and the fish yield in Aboket village.
 - (ii) the cause of the challenges experienced by the community.
- (b) Explain the measures the residents should take to reduce the challenges reported by the environment officer.

In the fast-growing town of Namanve, a large open-air market has become a significant source of environmental concern. Every week, tons of waste are discarded in an unprotected area behind the market stalls. This waste pile consists of a variety of materials that include; plastics old phones, batteries, old shoes, old clothes and rotting food materials. Mr. Mukasa, a local farmer whose garden is located directly downhill from this waste site, has observed a sharp decline in his crop productivity. His maize plants appear stunted with yellowing leaves (chlorosis), and the soil has become increasingly difficult to plough because of thick layers of buried plastic. During the rainy season, dark-coloured liquid (leachate) from the waste pile flows directly into his garden.

Task

- (a) Explain how the different categories of waste at the market site are contributing to the degradation of the soil structure and fertility in Mr. Mukasa's garden
- (b) Suggest sustainable waste management strategy for the Namanve market that would help the community achieve the Sustainable Development Goal of "Life on Land."

ERETA

In many urban settlements people eat roasted meat which sometimes never cooks well, and dispose of fecal matter poorly due to limited toilet/latrine facilities. Due to poor drainage, the rain water during heavy downpour wash the streets ending up into rivers and streams which are also sources of water for drinking, washing and feeding their animals. The public health officer has received reports of people suffering from stomach discomfort, weight loss, itchy anus, poor appetite, headaches and pass out stools with tissue fragments which seem to be of an organism.

Task.

- a) Explain the challenges faced by the residents of these urban settlements.
- b) How can these challenges be overcome?

Many parts of western Uganda are characterized by savanna grasslands where animals roam freely as they feed on grass, hunt and kill each other for food, and their body remains are broken down by other organisms. Wild dogs have been observed to have rough skins which are heavily attacked by blood sucking organisms. Residents in these lands use bushes for as latrine/toilet facilities, apply chemicals to kill termites which destroy their crops and property, and also cause fires to scare animals into traps during hunting. As a result, some animals are becoming rare, gardens crops are not yielding well and people are suffering illness but have no local medicines to use. This has worried health and wild life conservation officials of what may follow if the residents continued to act the same way.

Task.

- a) Explain
 - (i) the relationship that exists between the different organisms cited in the grasslands.
 - (ii) how is the behaviour of residents becoming a concern to the authorities?
- b) Measures to prevent the challenges that may arise due to the behaviour of the residents.

SEK

Lake Albert is an important source of fish, water, and transport for communities in Buliisa District. Over the years, the population around the lake has increased. Some fishermen use chemicals and small-mesh fishing nets to catch fish. Some households wash clothes, bathe, and dispose of rubbish directly into the lake. During the rainy season, fertilizers and pesticides from nearby gardens are washed into the lake. As a result, fish catches have reduced, bad smells are sometimes noticed in the lake water, and some people develop water-borne diseases after using the lake water.

Task

- (a) What are the possible effects of the activities of the residents of the Lake Albert?
- (b) Explain why such a natural resource should be well protected.

A wetland near a secondary school was drained to create land for more dormitory construction, crop growing and brick making. Trees around the wetland were cut for firewood. Within a few

years, floods became common during the rainy season, while water sources dried up during the dry season. Many birds, frogs and insects disappeared from the area.

Task.

- (a) Explain why draining wetlands caused the problems as in the scenario.
- (b) Suggest restoration measures that should be practiced to solve the rising issues.

ASSHU BUSHENYI

An agrochemical factory was established in a swamp which had a flowing river supporting community members with fresh water and mud fish. The factory produces a lot of smoke into the atmosphere and effluents into the swamp. After ten years, the water in the river turned blackish with a lot of legal blooms. The mud fish harvests reduced and many community members were diagnosed with respiratory diseases. The community members have recorded serious complaints with the local leaders.

Task

- a) Explain how agrochemical factory has affected the ecosystem in the area.
- b) Why should such ecosystem be conserved?

In 2018, the government of Uganda decided to give away part of Mabira Forest to Metha group of companies to extend sugar cane plantations. The agriculture-investor cleared the forest in the area, used tractors to plough the land and planted the sugar canes. For several years of extensive sugar cane growing, the investor applies artificial fertilizers for better yields of sugar canes. He often burns down the garden during harvest to obtain sugarcane for processing. After several years, the area now experiences reduced rainfall amounts and the investor complains of reduced sugar cane yields despite using fertilizers extensively.

Task

- a) Explain how the investor's farming practices have resulted into the challenges experienced in the area.
- b) How can the investor use the environment without much damage?

UTES

During the recent dry season, Asima set the dry grassland on fire and cut down all the trees in an attempt to clear a large portion of land for crop growing. After some time, people started complaining of destruction of their crops by strong winds, reduced catch by hunters, breathing difficulties, and attacks by wild animals.

Task

- (a) Explain how the challenges faced by the people were brought about by Isima's actions.
- (b) Why was it important for Mr. Asima to leave that area undisturbed?

JJEB

A farmer cleared a large area of natural vegetation to establish a crop field. He began growing maize intercropped with millet and, during the first three planting seasons, he obtained high yields. After several seasons of growing the same crops, the yields started to decline, and pest infestations became more severe. In response, the farmer began applying artificial fertilizers and

broad-spectrum pesticides every season. Although yields improved temporarily, they later declined again. He also noticed that the fruit trees growing near the crop field produced fewer fruits than before.

Task

- a) Explain how the activities of the farmer caused the observed challenges.
- b) Advise the farmer on how he can sustainably ensure high yields.

In the dry plains of a cattle corridor in Uganda, residents cut down trees for charcoal burning, burn grass during the dry season to control parasites and keep large herds of cattle. As a result, the hunters have observed a reduction in wild animals in the area, increased populations of rats in their homes and increased environmental temperature.

Task

- a) Explain how the challenges in the area were brought about by the activities of the residents.
- b) Why is it important for the residents to sustainably use the area?

KASSPHA

A farmer converted a large area of natural grassland into a commercial sugarcane plantation. He removed all native grasses and trees, ploughed the land deeply, and planted sugarcane in monoculture. He also uses large amounts of chemical fertilizers, herbicides, and pesticides. He diverted irrigation water from nearby streams. Recently, he noticed that the soil has become compacted, the water table has dropped, and the streams have become polluted. The farmer is also experiencing reduced sugarcane yields despite increased chemical use.

Tasks

- a) Explain to the farmer the environmental challenges he has caused.
- b) Explain why it is important for the farmer to adopt more sustainable farming practices.

A chemical factory was established on a large river bank, with a view that it needs constant supply of water for its machinery and activities. Overtime, the river developed a lot of algal blooms, and the fishermen complained of reducing numbers of fish species previously caught in large numbers. It was also noted that during the establishment of the factory, a lot of soil was heaped along the river bank, and so a lot of deposition goes into the river. Environmentalists have warned the government about pending dangers especially during rainy seasons.

Tasks

- a) Explain to the relevant authorities the origin of the problems
- b) Sensitize community on the relevancy of the maintaining such communities in their natural form

LUDISSEB

John acquired a piece of land near a community river to start a commercial goat farming project. The land was originally covered with natural forest. To prepare it, he cleared all the trees and used tractors to plough the area. He later burnt the cut trees to produce charcoal, which he sold to raise extra money for his project. After setting up the farm, the surrounding community began to experience challenges in their local environment.

Task

- (a) Explain the possible environmental challenges that may have resulted from John's activities on the land and the nearby community.
- (b) Suggest possible solutions to the challenges that would promote environment recovery

A civil war broke out in one of the hilly communities in Africa, during which wild animals were hunted for food, plantations, grasslands and people's homes were burnt down, and this produced thick smoke ever seen in the atmosphere. The land was left bare. After the war the survivors in the community tried to grow some crops to fight food insecurity but the yields were low and poor in quality. They also complain always about the snakes and other dangerous animals that attack them.

Task

- (a) Explain the environmental challenges facing the community
- (b) Suggest possible solutions to the challenges.

ITEM 6 AND 7

ACEITEKA

Mr. Zimbali took a cup of porridge before going for his garden work. He started digging with Vigor but later felt muscle pain and fatigue. When he became ill at one time, he went for medical checkup and was found with high blood pressure and arteriosclerosis. The doctor informed him that he needs a combination of management interventions to improve his health condition.

Task

- (a)
 - (i) Describe how the porridge enabled Mr. Zimbali carry out his exercises in the gym.
 - (ii) Explain the causes of the challenges his body experienced during digging and those discovered during medical checkup.
- (b) Suggest management interventions to improve Mr. Zimbali's health.

A patient with a habit of smoking had been diagnosed with coronary heart disease. During a medical checkup, a doctor used a stethoscope to find out how blood moves within his body pumping organ. Further medical tests revealed that his red blood cells had low oxygen loading, a reason why he was generally weak.

Task

- (a) Provide information to explain
 - (i) What the doctor wanted to find out during the medical examination.
 - (ii) How the patient's habit resulted into the health challenges diagnosed in him.
- (b) Suggest possible solutions to improve the patient's condition.

UTES

During first term holidays, Tom went back home to help her mother with garden work. One day as he started digging, he was energetic, breathing normally and having normal heart beat. As time went by, he became weaker, breathing faster, having faster heartbeat and feeling pain in his arms. He decided to rest but continued breathing deeply and the pain in the arms went on

reducing. Upon going back home, he was given a cup of millet porridge. Thirty minutes later, he became stronger.

Task

- (a) Explain the challenges Tom's body faced.
- (b) How were the challenges in his body overcome?

In an attempt to maintain her slender figure, Irene started avoiding having solid meals but would take dietary pills and sweets. She also smokes cigarettes with her friends. However, she has developed breathing difficulties, complains of toothache and was recently diagnosed with high blood.

Task

- (a) Explain how Irene's lifestyle caused her the health challenges.
- (b) What advice can you give to Irene to live a better life?

JJEB

A five-year-old boy in primary one, suffered nutritional deficiencies which presented the following signs and symptoms; bow-shaped legs, bleeding gums, pot belly and a larger body size for his age. His Physical Education teacher realized that he is extremely slow during physical exercises but encouraged him to continue in order to improve his health. One day, during exercise, the teacher observed that the boy's heartbeat and breathing rate increased so much and he was complaining of muscle pain. She advised him to rest but the boy continued breathing fast for a while before returning to the normal rate.

Task

- a) Explain the causes of the boy's health challenges.
- b) Explain the body changes that occurred during and after exercise.

A veterinary officer has been mixing his drugs while smoking cigarettes inside his poorly ventilated drug shop before going out to treat animals for his customers. He usually misses lunch since he spends the whole day attending to his various customers. Recently, he became ill. He was taken to hospital and the clinical laboratory report indicated that he has broken alveolar walls, very few cilia lining the air passages, wounds in the inner lining of the stomach and few villi in the ileum.

Task

- a) Explain how the officer's lifestyle and working conditions led to the observations in the laboratory report.
- b) Advise the veterinary officer on how he can improve his health and working conditions.

KASSPHA

Grace, of blood group B, was involved in a terrible boda-boda accident, sustained deep cuts, and lost a massive amount of blood. Due to the heavy bleeding, her blood volume and blood pressure dropped dangerously low. This drop in blood pressure immediately affected her kidneys, disrupting the process of ultrafiltration. To save her life, Grace's brain released ADH since her body could not afford to lose any more fluid. Because her body reabsorbed almost all the water, Grace produced only a tiny amount of very thick, dark urine. To save her, the doctors needed to

give her an emergency blood transfusion. Her relatives were confused about who could give her blood since the health center supplies were low.

Tasks:

- a) Describe the sequence of events that occurred in Grace's body from the moment of the accident to the production of thick, dark urine.
- b) Without referring Grace to any hospital, suggest with reasons which relatives with particular blood groups could donate blood to Grace and those who cannot.

Kefa, a teenager who had spent two years in the company of chain smokers, developed a persistent dry cough, chest pain, breathing difficulties, and eventually coughed up blood. After being admitted to hospital, doctors found that his alveoli had been severely damaged. Further examination revealed that Kefa was severely malnourished due to eating very small portions of a purely carbohydrate diet, resulting in low haemoglobin levels and constantly feeling weak.

Tasks

- a)
 - i) Explain the causes of Kefa's symptoms and poor health.
 - iii) Recommend and explain appropriate solutions to improve his health.
- b) Suggest with reasons, a diet for Kefa that will address his malnutrition and support recovery.

A five-year-old boy in primary one, suffered nutritional deficiencies which presented the following signs and symptoms; bow-shaped legs, bleeding gums, pot belly and a larger body size for his age. His Physical Education teacher realized that he is extremely slow during physical exercises but encouraged him to continue in order to improve his health. One day, during exercise, the teacher observed that the boy's heartbeat and breathing rate increased so much and he was complaining of muscle pain. She advised him to rest but the boy continued breathing fast for a while before returning to the normal rate.

Task

- a) Explain the causes of the boy's health challenges.
- b) Explain the body changes that occurred during and after exercise.

A veterinary officer has been mixing his drugs while smoking cigarettes inside his poorly ventilated drug shop before going out to treat animals for his customers. He usually misses lunch since he spends the whole day attending to his various customers. Recently, he became ill. He was taken to hospital and the clinical laboratory report indicated that he has broken alveolar walls, very few cilia lining the air passages, wounds in the inner lining of the stomach and few villi in the ileum.

Task

- a) Explain how the observations in the report led to the officer's ill health.
- b) Apart from treatment, what advice can you give to him to improve his health?

ASSHU

Joan has been smoking cigarettes, eating a lot of fatty meat and drinking fluids with high sugar content. When she went for medical examination, her BMI was found to be higher than normal. She had developed swollen feet, had breathing difficulty and was diagnosed with cardio-vascular disease. Joan's vision had also become blurred.

Task:

- (a) Justify Joan's health problems in relation to her lifestyle and feeding habits.
- (b) Apart from medical treatment, how can Joan improve her health conditions?

David's 3-year-old son was attacked by sickness that resulted in the breathing difficulties, high body temperature, vomiting and sores in the mouth and ulcers in the small intestines. The baby was rushed to the hospital and after some treatment, the breathing normalized and the vomiting ceased. The doctors advised that although the baby was still anaemic, the most appropriate treatment is proper diet that can enable the baby fully recover.

Task

- (a) Explain how the sickness affected the normal functioning of the baby's body.
- (b) Suggest an appropriate diet with reasons which will enable the baby to recover as per the Doctor's advice.

UTEC

Mr. Daniel Mwesigwa has spent the last 30 years working in a cigar-rolling factory in Mbale. The workshop is stuffy and poorly aired. Most of the staff members smoke on the job, so Daniel breathes in a lot of tobacco fumes every day. Because of overtime shifts, he usually eats only one meal a day, mostly cassava and black tea. Lately, Daniel develop a long time cough, feels worn out all the time and gets out of breath easily. At the hospital, tests showed he is severely anaemic, his bones are fragile, his gums bleed often. Doctors said he needs a blood transfusion urgently. Daniel is blood group O. His wife, Amina, who is blood group A, volunteered to donate. The medical team refused her blood, which made Amina angry and hurt.

Task

- a) Explain;
 - (i) To Amina, why she could not donate her blood to Daniel. And the anger to giving it.
 - (ii) How Daniel's job conditions and eating pattern have caused the health problems he is facing.
- b) Explain the urgent medical care and lifestyle adjustments Daniel should make to recover and bring his body back to normal function.

During the National Youth Skills Expo in Gulu, two students were picked for the 2-hour Furniture Making Challenge. The task required constant standing, cutting, and assembling wood. At the 60-minute halfway point, Ivan ate a large plate of posho and beef stew before continuing. Grace took a small bottle of mixed fruit juice with dates and honey, and then went back to work. In the last hour, Grace stayed focused and finished her table first. Ivan slowed down, complained of a full, heavy stomach and felt faint. Their habits during practice week were different: Ivan usually had one big, starchy meal right before long practice sessions. Grace ate a balanced diet with whole grains, fish, vegetables, fruits, and drank water regularly.

Task:

- a) Explain
 - (i) how the mouth and the pancreas helped in digesting Ivan's meal at the halfway point.
 - (ii) The reasons why Grace was able to complete the challenge better than Ivan.

- b) Suggest and explain changes Ivan should make to his diet and routine to improve his endurance and performance.

KAMSSA

Kasango is a businessman in Masaka town who spends his leisure time drinking alcohol and smoking cigarettes with his peers. He oftenly feeds on fatty meat and prefers eating foods with a lot of sugars from a local restaurant and thus added a lot of body weight. His doctor advised him to engage in regular physical exercises in order to keep him physically fit and to prevent him from suffering from some health-related problems. However, during his first visit to the gym, he experienced heavy breathing, increased heart rate, painful muscle cramps and sweated a lot.

Task

- (a) Explain the possible causes of Kasango's experience while exercising.
(b) Apart from engaging in physical exercise, how can you advise him to manage his lifestyle?

While watching last year's Olympic games on the television set, Cheptegei who represented Uganda in the games won the race and was awarded a medal. However, before the competitions began, he was given glucose powder by his coach as an energy booster. During the race, his heartbeat and sweating rate increased. He was also seen breathing deeply as he ran around the pitch with a lot of fatigue. Finally, he won the race and all these disappeared and was able to move to receive his medal and also celebrate his victory.

Task

- (a) Explain the challenges experienced by Cheptegei during the race.
(b) How Cheptegei was able to move to receive his medal, and how his body returned to a normal state after the race.

UMTA

While in the hospital, a diabetic man was requested to do a urinalysis test to assess the functioning of his kidney. The results below were extracted from the laboratory report of the hospital.

Test Specimen	Test type	Observation	Test status
Urine	Benedict's test	Solution turned brown	Positive test
Urine	Biurette test	Solution turned purple	Positive test

Task:

- (a) Explain how,
(i) the components detected in the man's urine are caused by a malfunction of the assessed organ?
(ii) high amounts of the components detected may affect normal body functioning?
(b) Suggest possible remedies to improve man's condition.

Ahmed started working during his fasting period at a cement factory that produces a lot of dust and smoke. He always feels dizzy at work and after a while he began experiencing some chest pain and other body complications that made him ineffective at his job. When he visited the hospital, the doctor told him that he was suffering from emphysema, high blood pressure, hypoglycaemia and muscle wasting.

Task:

- (a) Explain how Ahmed's Health complications were as a result of his work and lifestyle?
- (b) Advise Ahmed how he can handle the health challenges.

LUDISSEB

Mr. Kato's behaviour of smoking and drinking alcohol has caused swelling of his lungs, blockage of pancreatic ducts and damage of the stomach lining (ulcers). He persistently feels general body weakness and fatigue.

Task

- (a) Describe how the effects of Kato's behaviour affected the general functioning of his body.
- (b) Advise Kato on how to manage his health conditions.

In an attempt to treat leukemia, Peter a 17-year-old underwent a chemotherapy procedure which suppressed production of blood cells, weakened his heart muscles and reduced his appetite due to mouth sores (wound). He later started experiencing fatigue during exercise and frequent illness.

Task

- (a) Explain how the side effects of chemotherapy affected the normal functioning of his body leading to frequent illness and fatigue.
- (b) How can Peter manage the side effects experienced because of the procedure?

WAKISSHA

Sarah a 16-year-old student, for several weeks experienced frequent urination, excessive thirst, tiredness, blurred vision and unexplained weight loss. One morning at school, she fainted during assembly and was rushed to a nearby health centre. The health worker suspected diabetes and tested Sarah's blood glucose level before and after she ate a meal. Her urine was also tested for glucose and proteins.

Table 1 shows Sarah's Blood glucose levels.

Time	Blood glucose levels mmol/l
Before meal	12.5
1 hour after meal	19.8
2 hours after meal	17.6
4 hours after meal	14.2

Urine test results showed absence of proteins and presence of glucose; and the normal fasting glucose level is 4-6 mmol/l.

Task

- (a)
 - (i) Using the information in the scenario and data in the table 1, explain why the health worker suspected diabetes.
 - (ii) Explain how diabetes affected the functioning of Sarah's body.
- (b) Suggest suitable ways Sarah can manage her condition without being referred to another hospital and explain what may happen to Sarah if the condition is NOT managed properly.

Chemongesi is a talented long-distance runner, who smokes regularly. Their home is located near a swampy area. Over the past six months, Chemongesi started experiencing frequent

illnesses. Medical tests later confirmed that he has developed heart and respiratory complications, alongside recurrent malaria infections. His Coach noticed a steady decline in his athletic performance and collected the following data;

Table 2: Chemongesi's monthly performance.

Month	Distance covered	Malaria episodes (number of times suffered)	Body weight (kg)	Lung function lung volume(cm ³)
January	6.5	0	60	420
February	6.2	1	59	400
March	5.8	2	57	370
April	5.1	2	55	340
May	4.3	3	53	310
June	3.5	3	50	280

Task

- Using the data provided, explain the decline in Chemo Ngesi's sports performance.
- Suggest measures Chemongesi can take to manage his health condition and improve his-wellbeing.

ERETA

A renowned kickboxer on a school's kickboxing club team suddenly developed fatigue, pain in the lower back, seizures, shortness of breath, increased pulse rate and passing out little urine with crystalline solids. Following a medical examination and investigations, it was reported that he cooked using charcoal in closed rooms, ate a lot of salty red meat, barely drunk enough water across the day and his major body organs had been damaged.

Task.

- Identify the organs that could have been damaged leading to the conditions experienced by the kickboxer.
- What should be done by the kickboxer to manage the conditions he experienced?

In a rural trading centre, Mr. Okello earns a living by producing and selling charcoal. He spends many hours cutting trees, burning wood in charcoal kilns, carrying sacks of charcoal and standing in smoky environment. He often skips meals while working, drinking very little water and rarely wears protective gear. After several months, he developed persistent cough, breathlessness during work, fatigue, muscle weakness, passes dark urine on hot days, and occasional dizziness. A health worker advised him to improve his lifestyle and medical care.

Task.

- Explain the causes of Mr. Okello's health problems.
- Suggest a health improvement plan for Mr. Okello.

SEK

An NGO organized a three-day learner's health and sports camp. During the camp, learners participated in football, athletics and long-distance running. The organization provided balanced meals containing posho, beans, rice, meat, vegetables, fruits and clean drinking water. However, a few learners skipped breakfast, drank only soft drinks instead of water and spent most of their free time resting instead of joining physical activities. During the final race, these learners became

tired very quickly, breathed heavily and later complained of muscle pains. A nurse who was hired for the entire camp activities also discovered that some of them passed very little urine that was dark yellow in colour.

Task.

- (a) Explain the difference in performances of the two categories of the learners who participated in the activities of the health and sports camp.
- (b) Account for the learners' breathing rate, muscle pains experienced and the colour of the urine passed.

A group of tourists visited a National Park. They walked for several kilometres under the hot sun. Before leaving for the traverse, tour guides advised them to eat breakfast, carry drinking water and avoid unhealthy snacks. Some tourists followed the advice while others carried only sweets and soft drinks. As the journey continued, those who ignored the advice became weak, thirsty and exhausted. They climbed higher on a hill, but realized that their breathing rate all of a sudden increased. They also sweated heavily and later produced only a small amount of concentrated urine.

Task

- (a) Explain the relevance of the tour guides advise to the tourists.
- (b) Explain the observations made as the tourists continued with their journey.

KEEBS

St. Mark Secondary School recently organized a "Health and Wellness Fair" to sensitize the surrounding community about common lifestyle diseases. As part of the Biology project, the S4 students were tasked to demonstrate how the body processes nutrients and why health screenings specifically urine analysis are critical for early detection of metabolic imbalances. During demonstration, a 45-year-old local shopkeeper, Mr. Opio, who complained of persistent fatigue, frequent thirst after the meal of sweet potatoes and eggs, also complains of rapid weight loss. Provided his medical report papers to students to explain to him. On the report, the following lab results were written

Lab report

Test	Results/observation
Benedict's test on urine (presence of glucose)	Turned brick-red
Biuret test (presence of proteins)	Turned purple

The students then had to explain to the local residents not only how these substances reached the urine but also the deeper physiological implications of their presence in a person's excretory system.

Task

- (a) Explain to the residents how nutrients in Mr. Opio's meal reached urine
- (b) Explained to Mr. Opio the cause of what he is experiencing and advise him on how to manage the conditions

Mr. Balu is a 50-year-old taxi driver operating in the busy, dusty streets of Kampala. To cope with the long, stressful hours, he smokes two packets of cigarettes daily. His diet consists mostly of "Rolex" (fried chapatti and eggs) and deep-fried chicken from roadside stalls, which he washes

down with sugary sodas. For the past six months, Mr. Balu has developed a persistent, "wet" cough that refuses to go away. Despite his high-calorie diet, he is noticeably losing weight and feels constantly weak. Recently, he started experiencing sharp chest pains and breathlessness whenever he climbs the stairs to his apartment.

Task

- (a) Explain the cause of Balu's health conditions
- (b) Suggest the possible ways Balu can manage his health status

SEEBS

Paul a 20-year-old resident of Mayuge district is one of the selected athletes who are to represent the district in the regional 10-km race competitions. In the past 2 years, Paul has been a heavy cigarette smoker who rarely trains and depends on fat rich diet which is deficient in iron. During the race Paul experienced extreme tiredness and fell down and was rushed to the hospital. In the hospital the doctor reported that Paul of blood group B needed blood transfusion.

Tasks:

- a) Show how Paul's habits in the past 2 years led to the challenge of extreme tiredness during the competitions.
- b) (i) Basing on his habits, propose strategies he can undertake to improve his performances.
(ii) With a reason in each case, suggest possible blood types that can be administered to Paul by the Doctor.

Mr. Kato a 45-year-old man visited a health centre after experiencing swollen legs and extreme tiredness. The doctor suspected his kidneys to be malfunctioning. The doctor obtained a urine sample from Mr. Kato and below are the laboratory results.

Substance	Urine of healthy person (%)	Mr. Kato's urine (%)
Water	95	92
Salts	1.5	3.2
Red blood cells	0	0.8
Urea	2.0	2.8

The doctor later advised Mr. Kato to eat little salt, very little meat, take little water and if possible, get a kidney transplant from a donor of compatible blood type.

Tasks:

- a) (i) Suggest reasons for the difference in the composition of Mr. Kato's urine compared to the normal urine
(iii) Using data in the table, show how Mr. Kato's symptoms have been brought
- b) Justify each point in the doctor's advice.